

Design Thinking (AECC27)

Sample Questions and Solutions

UNIT-I: Introduction

Q1. Define design thinking and give its importance.

Design thinking is a problem-solving approach that places human needs and experiences at the centre of the design process. It is a mindset and a set of methods and tools that empowers individuals and teams to approach complex challenges in a creative, iterative, and user-centric manner.

Design thinking typically follows a structured process that involves the following stages:

- i. ***Empathize:*** Understanding the needs and perspectives of the people for whom you are designing. This involves engaging in research, conducting interviews, and observing user behaviour to gain deep insights into their desires, challenges, and motivations.
- ii. ***Define:*** Synthesize the information gathered during the empathy stage to identify the core problem or opportunity that needs to be addressed. This involves reframing the challenge into a clear problem statement that guides the design process.
- iii. ***Ideate:*** Generating a wide range of creative ideas to solve the defined problem. This stage encourages brainstorming, thinking outside the box, and considering multiple perspectives. The focus is on quantity and suspending judgment.
- iv. ***Prototype:*** Building tangible representations of the ideas generated during the ideation stage. Prototypes can take various forms, such as physical models, sketches, storyboards, or even interactive digital simulations. The goal is to quickly bring ideas to life to gather feedback and iterate.
- v. ***Test:*** Gather feedback on the prototypes from the intended users and stakeholders. This stage involves conducting user testing and evaluation to understand how well the design solutions meet the users' needs. Iteration and refinement are crucial at this stage.

The importance of design thinking lies in its ability to:

- i. ***Enhance User-Centered Solutions:*** Design thinking places users at the core of the design process, ensuring that solutions meet their needs, desires, and preferences. By empathizing with users, designers gain deep insights that lead to more relevant and effective solutions.
- ii. ***Foster Creativity and Innovation:*** Design thinking encourages a divergent and exploratory mindset, enabling designers to think outside the box and generate novel ideas. It promotes an environment that nurtures creativity and sparks innovation.
- iii. ***Drive Human-Centric Design:*** Design thinking shifts the focus from technology or business constraints to human needs. It promotes a holistic understanding of users and their context, enabling designers to create products and services that resonate with their emotions, aspirations, and behaviours.
- iv. ***Facilitate Collaboration and Interdisciplinary Teams:*** Design thinking emphasizes cross-disciplinary collaboration by bringing together individuals with diverse backgrounds, skills, and perspectives. This collaborative approach leads to more comprehensive and innovative solutions that draw from multiple fields of expertise.
- v. ***Enable Iterative and Adaptive Problem-Solving:*** Design thinking embraces an iterative process of prototyping, testing, and iteration. This approach allows designers to gather

feedback early on, learn from failures, and make continuous improvements, reducing the risk of developing solutions that miss the mark.

- vi. **Encourage Empathy and Human-Centered Research:** Design thinking places a strong emphasis on understanding and empathizing with users. Through methods like user interviews, observation, and ethnographic research, designers gain deep insights into user needs, pain points, and motivations, leading to more meaningful and impactful solutions.
 - vii. **Enhance User Experience and Satisfaction:** By prioritizing user needs and involving users throughout the design process, design thinking helps create products and services that deliver exceptional user experiences. This leads to higher user satisfaction, loyalty, and positive brand perception.
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Q2. What is Design Thinking? How does design thinking work?

Design thinking is a problem-solving approach that combines empathy for the user, creativity, and rationality to generate innovative solutions. It is a human-centric and iterative process that seeks to understand user needs, redefine problems, and develop and test multiple ideas to arrive at effective solutions. Design thinking involves the following stages:

- i. **Empathize:** The first step in design thinking is to empathize with the users and gain a deep understanding of their needs, desires, and challenges. This stage involves conducting research, interviews, and observations to gather insights about the users and their experiences.
- ii. **Define:** In this stage, the information gathered during the empathy stage is synthesized to define the core problem or challenge that needs to be addressed. Designers reframe the problem statement based on user insights, ensuring a clear understanding of what they are trying to solve.
- iii. **Ideate:** During the ideation stage, designers generate a wide range of creative ideas and potential solutions. The emphasis is on divergent thinking and suspending judgment. Brainstorming sessions, sketching, mind mapping, and other techniques are used to encourage creativity and generate as many ideas as possible.
- iv. **Prototype:** In this stage, designers create tangible representations or prototypes of their ideas. Prototypes can take different forms depending on the context, such as physical models, digital simulations, storyboards, or even role-playing. Prototyping allows designers to visualize and communicate their ideas and make them tangible for further exploration and testing.
- v. **Test:** The testing stage involves gathering feedback on the prototypes from users and stakeholders. Designers observe how users interact with the prototypes, gather their feedback and insights, and use that information to refine and improve the solutions. Iterative cycles of testing and refinement help designers identify strengths, weaknesses, and areas for improvement.

Design thinking is characterized by its iterative nature, which means that the process often loops back to previous stages as new insights and feedback are incorporated. The stages of design thinking are not necessarily sequential and can be revisited multiple times to refine the solutions. This iterative process allows for continuous learning, adaptation, and improvement.

Throughout the design thinking process, collaboration and interdisciplinary teamwork are essential. Design thinkers bring together individuals with diverse backgrounds, skills, and perspectives to generate a wide range of ideas and insights. This collaborative approach fosters innovation and ensures that solutions are comprehensive and effective.

Design thinking can be applied to a wide range of challenges, from product design and service development to organizational processes and social innovation. It provides a structured framework that encourages a human-centred, creative, and iterative approach to problem-solving, resulting in solutions that address real user needs and deliver meaningful experiences.

Q3. Why companies need to apply design thinking.

Companies can benefit from applying design thinking for several reasons:

- i. ***Customer-Centric Solutions:*** Design thinking places the customer at the heart of the design process. By understanding and empathizing with customer needs, companies can create products, services, and experiences that truly resonate with their target audience. This customer-centric approach leads to greater customer satisfaction, loyalty, and advocacy.
- ii. ***Innovation and Differentiation:*** Design thinking fosters a creative and exploratory mindset, encouraging teams to think beyond conventional solutions. By generating a wide range of ideas and prototyping them, companies can uncover innovative and unique approaches to challenges. This can help them stand out in competitive markets and differentiate their offerings.
- iii. ***Problem-Solving Agility:*** Design thinking promotes an iterative and adaptive problem-solving approach. By rapidly prototyping and testing solutions, companies can identify and address potential issues and risks early on. This iterative process allows for course correction, reducing the time and cost associated with developing ineffective or undesirable solutions.
- iv. ***Collaboration and Interdisciplinary Teams:*** Design thinking encourages cross-functional collaboration and the inclusion of diverse perspectives. By involving individuals from different backgrounds and areas of expertise, companies can benefit from a collective intelligence that leads to comprehensive and well-rounded solutions. Collaboration also fosters a sense of ownership, engagement, and shared purpose among team members.
- v. ***User Experience and Brand Perception:*** Applying design thinking results in products and services that deliver exceptional user experiences. This positive user experience not only increases customer satisfaction but also enhances the company's brand perception. Designed companies are often associated with quality, innovation, and user-centricity, which can positively impact customer loyalty and brand equity.
- vi. ***Empathy and Insights:*** Design thinking emphasizes understanding and empathizing with customers. Through research, interviews, and observations, companies gain deep insights into customer needs, pain points, and motivations. This customer understanding helps companies create solutions that are tailored to their target audience, ensuring a better fit with their preferences and requirements.
- vii. ***Agility and Adaptability:*** Design thinking promotes a mindset of experimentation and learning from failures. By encouraging a willingness to take risks, companies can quickly

adapt and respond to changing market dynamics and customer needs. This agility allows them to stay ahead of the competition and seize new opportunities.

Q4. Reasoning out the importance of the 4C's namely critical thinking, creative thinking, communicating and collaborating is an essential skill set for students with suitable examples.

The 4C's - Critical Thinking, Creative Thinking, Communicating, and Collaborating - are essential skill sets for students because they empower them to become effective problem solvers, innovative thinkers, and competent team players. These skills are crucial for success in academics, careers, and life in general. Let's explore the importance of each of these skills with suitable examples:

i. Critical Thinking:

Importance: Critical thinking enables students to analyze information, evaluate arguments, and make reasoned judgments. It helps them develop a logical and rational approach to problem-solving, enabling them to distinguish between reliable and misleading information.

Example: When researching a topic for a school project, critical thinking skills allow students to assess the credibility of sources, identify biases, and evaluate the validity of arguments. This ensures that they rely on accurate and trustworthy information to support their work.

ii. Creative Thinking:

Importance: Creative thinking fosters innovation, imagination, and the ability to generate new ideas. It encourages students to think outside the box, explore multiple perspectives, and find unique solutions to challenges.

Example: In a science class, creative thinking skills empower students to propose alternative hypotheses, design creative experiments, and develop innovative solutions. They might devise new ways to solve environmental problems, develop inventions, or approach scientific inquiries from unconventional angles.

iii. Communicating:

Importance: Effective communication skills are essential for expressing ideas, sharing information, and connecting with others. Students with strong communication skills can articulate their thoughts clearly, actively listen to others, and adapt their communication style to different audiences.

Example: During a class presentation, effective communication skills enable students to organize their ideas coherently, use appropriate language and visuals, and engage their audience. They can convey complex concepts in a clear and concise manner, effectively delivering their message and fostering understanding.

iv. Collaborating:

Importance: Collaboration involves working together towards a common goal, sharing responsibilities, and valuing diverse perspectives. It promotes teamwork, empathy, and the ability to work effectively in groups.

Example: In group projects, collaborating skills enable students to delegate tasks, actively listen to their peers' ideas, incorporate different viewpoints, and find consensus. They learn to leverage the strengths of each team member, resolve conflicts constructively, and produce high-quality work collectively.

By cultivating the 4C's, students become well-rounded individuals equipped with essential skills for their personal, academic, and professional development. These skills enable them to approach challenges critically, come up with innovative solutions, communicate effectively with others, and collaborate successfully in various settings. As students transition into higher education and the workforce, the 4C provides a strong foundation for lifelong learning, adaptability, and success in a rapidly changing world.

Q5. Write your understanding of the four methods applied to design thinking.

Design thinking is a problem-solving approach that involves understanding and addressing user needs through a human-centred design process. There are several methods commonly applied within the framework of design thinking. Here is an overview of four key methods:

1. Empathize: The empathize stage involves gaining a deep understanding of users and their needs, motivations, and challenges. Methods used in this stage include:

- *User Interviews:* Conducting one-on-one interviews with users to gather insights and understand their experiences and perspectives.
- *Observation:* Observing users in their natural environment to uncover behaviours, pain points, and unmet needs.
- *Empathy Mapping:* Creating visual representations that capture users' thoughts, emotions, and behaviours to gain empathy and identify key insights.

2. Define: In the define stage, the insights gathered from the empathize stage are synthesized to define the problem statement and establish the design challenge. Methods used in this stage include:

- *User Persona:* Creating fictional representations of typical users to humanize and understand their needs and goals.
- *Problem Statement:* Framing the design challenge by identifying the specific problem to be solved and the desired outcomes.
- *Point of View (POV) Statements:* Formulating statements that articulate the user's needs, insights, and aspirations to guide the ideation process.

3. Ideate: In the ideate stage, creative solutions are generated to address the defined problem. This stage encourages brainstorming and thinking beyond conventional approaches. Methods used in this stage include:

- *Brainstorming:* Encouraging free-flowing idea generation without judgment, allowing for a broad range of potential solutions.

- *Mind Mapping*: Creating visual diagrams to explore and organize ideas, concepts, and connections.
- *Crazy 8s*: Providing a time-constrained activity to generate quick and diverse ideas through sketching or rapid prototyping.

4. *Prototype*: The prototype stage involves transforming selected ideas into tangible representations that can be tested and evaluated. Methods used in this stage include:

- *Paper Prototyping*: Creating low-fidelity prototypes using paper or other simple materials to quickly visualize and iterate on ideas.
- *Digital Prototyping*: Using digital tools to create interactive prototypes that simulate the functionality and user experience of a product or service.
- *Wizard of Oz*: Simulating certain aspects of a product or service to test and gather user feedback without fully developing the actual functionality.

It's important to note that these methods are not strictly linear and can overlap and iterate throughout the design thinking process. The methods applied in design thinking provide a structured framework for understanding user needs, defining problems, generating ideas, and prototyping solutions. This iterative and user-centered approach helps designers and teams create innovative and impactful solutions to complex problems.

Q6. If a designer needs to work with the full potential towards design, what would be the key mindsets required to achieve it? Analyze why and how.

A mindset is a set of attitudes that reflect how a person thinks or feels about a particular thing. These attitudes can affect the way they behave in a particular scenario. For example, if a student has a fear of failure and never wants to make mistakes, then he or she would be likely to hesitate in taking action, making important decisions, and trying new things when working in the DT process.

According to the K12 Lab Network wiki, six key mindsets are required by designers to use DT to its fullest potential:

1. Human centred
2. Mindful of process
3. Culture of prototyping
4. Bias toward action
5. Show don't tell
6. Radical collaboration

1. *Human-centered*: This mindset emphasizes the importance of understanding and empathizing with the needs, desires, and behaviors of the people who will be using or affected by a product, service, or system. Design thinking starts by putting humans at the centre of the problem-solving process. By observing and engaging with users, designers gain valuable insights that inform the development of innovative and effective solutions.

2. *Mindful of process*: Design thinking recognizes the significance of having a structured and iterative approach to problem-solving. It involves understanding and following a defined

process that typically includes stages like empathize, define, ideate, prototype, and test. Being mindful of the process means being aware of the specific actions and methods to be employed at each stage, and being open to revisiting and refining previous stages as new information is discovered.

3. Culture of prototyping: This mindset encourages designers to create tangible representations of their ideas early and often. Prototyping involves building low-fidelity or high-fidelity models, sketches, or simulations that allow designers to gather feedback, learn from failures, and iterate on their designs. It promotes a culture that values experimentation, risk-taking, and learning by doing, rather than relying solely on theoretical or abstract thinking.

4. Bias toward action: Design thinking advocates for taking action and making progress rather than getting stuck in analysis paralysis. Instead of endless discussions or planning, this mindset encourages a bias toward taking tangible steps forward. By embracing experimentation and learning through action, designers can gather real-world insights and make informed decisions. This bias toward action helps teams move quickly, test assumptions, and adapt their approach as needed.

5. Show don't tell: This mindset emphasizes the importance of visual and experiential communication over verbal or written explanations. Designers use visual aids such as sketches, diagrams, and prototypes to convey ideas, concepts, and solutions. By making ideas tangible and visible, designers can effectively communicate complex concepts, gather feedback, and align stakeholders. This mindset promotes a more engaging and inclusive way of communicating that encourages collaboration and understanding.

6. Radical collaboration: Design thinking promotes a collaborative approach that involves diverse perspectives and expertise. It recognizes that solutions to complex problems often require input from multiple disciplines and stakeholders. Radical collaboration involves actively seeking out different viewpoints, encouraging participation from various team members, and fostering an inclusive environment where everyone's contributions are valued. This mindset acknowledges that the best ideas can come from anywhere and that teamwork and shared ownership are crucial to successful outcomes.

Q7. Given below are the organizations

(i) NETFLIX

(ii) UBER,

justify how two well-known brands have leveraged design thinking to solve business problems.

(i) Netflix:

Netflix has effectively leveraged design thinking to solve various business problems and differentiate itself in the streaming industry. Here are some examples:

1. User-Centered Design: Netflix has adopted a human-centered approach, investing in understanding its users' preferences and behaviors. Through data analysis and user research, they gather insights into what content users enjoy, how they navigate the platform, and what influences their viewing choices. This deep understanding of user needs allows Netflix to curate personalized recommendations, tailor content offerings, and improve the overall user experience.

2. Iterative Prototyping: Netflix has embraced a culture of prototyping and testing. They continuously experiment with different user interfaces, content layouts, and recommendation algorithms. By creating prototypes and testing them with a subset of users, Netflix collects feedback and iterates on their designs. This iterative process helps them refine their product, enhance user engagement, and stay ahead of competitors.

3. Data-Driven Decision Making: Design thinking is complemented by Netflix's strong reliance on data analysis. The company collects vast amounts of data on user behaviour, content consumption patterns, and viewer feedback. They leverage this data to inform their decision-making processes, such as greenlighting new original series, renewing existing shows, or acquiring content. By combining design thinking with data-driven insights, Netflix maximizes its chances of delivering content that resonates with its user base.

(ii) Uber:

Uber has applied design thinking to revolutionize the transportation industry, solving several business challenges along the way. Here's how Uber has leveraged design thinking:

1. User-Centric Experience: Uber has fundamentally transformed the transportation experience by putting the user at the center. Through research and feedback loops, Uber has optimized the user journey from requesting a ride to arriving at the destination. The intuitive and user-friendly mobile app simplifies the process of hailing a ride, tracking the driver's location, and making cashless payments. This focus on the user experience has been instrumental in Uber's success.

2. Empathy with Drivers: Uber's design thinking approach extends beyond just the riders. The company recognizes the importance of empathizing with its driver-partners and understanding their pain points. By actively engaging with drivers and gathering feedback, Uber has introduced features like flexible work hours, driver ratings, and surge pricing, which address driver concerns and improve their experience. This focus on driver satisfaction helps Uber maintain a strong network of drivers and ensure the availability of rides.

3. Continuous Iteration: Uber embraces an iterative mindset, constantly refining its services based on user feedback and changing market demands. The company collects user ratings, reviews, and trip data to identify areas for improvement. Whether it's refining the app interface, enhancing safety features, or launching new service offerings like Uber Eats, Uber's iterative approach allows them to adapt and evolve their business model to meet evolving user needs.

By applying design thinking principles, both Netflix and Uber have effectively solved business problems, enhanced user experiences, and gained a competitive edge in their respective industries. They have demonstrated how human-centred design, iterative prototyping, and a focus on user feedback can drive innovation and success.

Q8. Explain the use of design thinking for business using any case studies or examples.

Design thinking is a problem-solving approach that is widely used in business to drive innovation, enhance customer experience, and create valuable solutions. It involves understanding the needs of users, generating creative ideas, prototyping and testing, and

iterating based on feedback. Here are a few case studies that demonstrate the use of design thinking in business:

1. Airbnb:

Airbnb, an online marketplace for lodging and vacation rentals, applied design thinking to transform the way people find and book accommodations. They focused on understanding the needs and desires of travelers and hosts through in-depth user research. By empathizing with their target users, Airbnb identified pain points and created a platform that provided unique and personalized travel experiences. This user-centered approach led to the design of features such as detailed property descriptions, user reviews, and a messaging system that facilitated trust and connection between hosts and guests. Through design thinking, Airbnb disrupted the traditional hospitality industry and became a global leader in the sharing economy.

2. IBM:

IBM used design thinking to drive innovation and transform its business approach. The company established IBM Design, a program that integrated design thinking into its culture and processes. Design thinking methods were employed to understand and address the complex challenges faced by IBM's clients. One notable example is the design transformation of IBM's enterprise software, where they shifted from a feature-focused approach to a user-centered one. IBM designers conducted extensive research to understand users' pain points and goals. Through iterative prototyping and testing, they developed a more intuitive and user-friendly software interface. This design-led approach not only improved the user experience but also helped IBM differentiate its offerings in the competitive enterprise software market.

3. Apple:

Apple is known for its human-centered design thinking approach, which has been instrumental in the success of its products. For instance, the development of the iPod and iTunes ecosystem involved understanding the needs and frustrations of music enthusiasts who struggled with managing their digital music libraries. Apple's design team conducted extensive user research, prototyping, and testing to create a seamless and intuitive user experience. They focused on simplifying the process of discovering, purchasing, and organizing music, resulting in a breakthrough product that revolutionized the music industry.

Q9. Illustrate “Human-centered” and “Show don’t tell” mindsets with examples.

i. Human-centred mindset:

A human-centred mindset in design means placing the needs, behaviours, and experiences of users at the forefront of the design process. It involves empathizing with users, understanding their context, and designing solutions that align with their goals and aspirations.

Example-1: When designing a new mobile banking app, a team with a human-centred mindset would conduct user research to understand the pain points and preferences of potential users. They might interview users, observe their interactions with existing banking apps, and gather feedback. This research would inform the design decisions, such as creating an intuitive user

interface, simplifying the navigation, and offering personalized financial insights to meet the users' needs and improve their overall experience.

Example-2: Imagine a design team working on a healthcare app for elderly users. With a human-centred mindset, they would conduct user research, such as interviews and observations, to understand the specific challenges and needs of older adults when it comes to managing their health. They might uncover issues like small font sizes, complex navigation, and difficulties in inputting data. Based on these insights, the team would then design an interface that features larger text, simplified navigation, and voice input capabilities, ensuring the app is user-friendly and accessible to its intended audience.

Example-3: Example: Let's say a team is designing a website for an online learning platform. With a human-centred mindset, they would conduct user research to understand the needs and preferences of their target users, such as students. They might conduct interviews, surveys, or usability tests to gather insights. Based on these findings, the team would create an intuitive interface, clear navigation, and personalized learning recommendations that cater to the specific needs and learning styles of the users. The design would prioritize ease of use, accessibility, and an engaging learning experience.

ii. Show don't tell mindset:

The "show don't tell" mindset encourages designers to communicate ideas, concepts, and information visually and tangibly, rather than relying solely on verbal or written explanations. It emphasizes the power of visual communication to convey messages, evoke emotions, and engage users.

Example-1: Instead of explaining the features and benefits of a new car model through a lengthy text-based brochure, a design team with a "show don't tell" mindset might create an interactive virtual reality experience. Users can put on a VR headset and explore the car's interior, test its advanced safety features through simulated scenarios, and take a virtual test drive. This immersive experience allows potential buyers to engage directly with the product, understand its unique selling points, and make a more informed decision based on their firsthand experience.

Example-2: Consider a marketing campaign for a new environmentally friendly product. Instead of simply telling consumers about the product's sustainability features through text-based advertisements, a show don't tell approach would involve creating visually compelling content. This could include impactful imagery depicting the product's positive impact on the environment, before-and-after visuals showcasing the difference it makes, or even interactive displays allowing consumers to experience the product's eco-friendly features firsthand. By showing rather than telling, the campaign engages consumers on a visual and emotional level, making the product more compelling and memorable.

Example-3: Imagine a product launch event for a new smartphone. Instead of presenting a series of slides with bullet points and text-heavy explanations about the phone's features, a

show don't tell approach would involve creating an immersive and interactive demonstration. The event may include live product demonstrations, where the audience can experience the phone's unique features, explore its capabilities through hands-on activities, and witness its performance in real-time. The event may also incorporate visual displays, such as videos or animations, to showcase the phone's sleek design and innovative functionalities. By letting the audience see and experience the product firsthand, the show don't tell approach creates a more memorable and engaging experience.

Q10. What does it mean to have a human-centred mindset? Differentiate it from a business/manufacturing-centred mindset.

Having a human-centred mindset means placing the needs, desires, and experiences of people at the core of the design process. It involves empathizing with users, understanding their behaviours, motivations, and pain points, and using that knowledge to create solutions that genuinely meet their needs. Here's how a human-centred mindset differs from a business/manufacturing-centred mindset:

i. Focus on User Needs: A human-centred mindset prioritizes understanding and addressing user needs. It begins by conducting research, gathering insights, and actively listening to users to uncover their challenges and aspirations. The focus is on creating solutions that truly serve and delight users, ensuring that their experiences are positive and meaningful. In contrast, a business/manufacturing-centred mindset may primarily emphasize operational efficiency, profit margins, and meeting business goals, potentially overlooking users' specific needs and desires.

ii. Empathy and Understanding: A human-centered mindset seeks to empathize with users and gain a deep understanding of their perspectives. It involves observing, interviewing, and engaging with users to gain insights into their experiences, emotions, and motivations. This empathetic understanding helps designers create solutions that resonate with users on a personal level. In contrast, a business/manufacturing-centered mindset may focus more on market demand, cost optimization, and production efficiency, often relying on assumptions and generalizations rather than direct user insights.

iii. Iterative and User-Driven Design: A human-centered mindset embraces an iterative design process, where prototypes and solutions are continuously refined based on user feedback and testing. It involves actively involving users in the design process, co-creating and co-designing solutions with them. This approach ensures that the end result is well-aligned with user needs and preferences. On the other hand, a business/manufacturing-centered mindset may follow a linear process focused on delivering predefined solutions efficiently, with limited room for user input or iteration.

iv. Emphasis on User Experience: A human-centered mindset recognizes the importance of user experience as a differentiating factor and a driver of success. It considers aspects such as usability, aesthetics, emotional connection, and overall satisfaction with the product or service. The goal is to create experiences that delight users and foster long-term engagement. In contrast, a business/manufacturing-centered mindset may prioritize cost reduction, mass production, and operational effectiveness, potentially overlooking the nuances of user experience.

In summary, a human-centered mindset is centered around the needs and experiences of users, emphasizing empathy, understanding, iterative design, and a focus on user experience. It differs from a business/manufacturing-centred mindset that may prioritize business objectives, efficiency, and profit, potentially neglecting the user's perspective. While both mindsets are important, adopting a human-centered mindset ensures that design solutions truly meet user needs and create meaningful experiences.

Q11. Why is it important to have a show-don't-tell mindset? How does it help the person in the design thinking process?

Having a "show-don't-tell" mindset is crucial in the design thinking process because it emphasizes the importance of tangible and visual representations over verbal or written explanations. It involves using prototypes, visual aids, and physical manifestations to communicate ideas, concepts, and solutions. Here's why it is important and how it helps individuals in the design thinking process:

i. Enhanced Understanding: Showcasing ideas through visual and tangible means helps foster a deeper understanding among team members and stakeholders. It allows them to grasp concepts more easily and effectively than relying solely on verbal descriptions or abstract explanations. When ideas are made tangible, it becomes easier to identify potential issues, generate feedback, and refine the design.

ii. Improved Communication: Visual and physical representations have the power to transcend language barriers and promote effective communication. They facilitate clearer and more concise communication by eliminating ambiguity and misunderstandings that can arise from verbal explanations alone. Showcasing ideas visually enables individuals to express complex concepts and intentions in a way that is easily comprehensible to others.

iii. Encourages Collaboration: By adopting a show-don't-tell mindset, individuals in the design thinking process can foster a collaborative environment. Visual representations serve as a common language that enables team members to share and build upon ideas more effectively. It encourages collaboration, as everyone can actively participate, contribute their perspectives, and engage in meaningful discussions based on shared understanding.

iv. Iterative Development: Showcasing prototypes and visual representations allows for iterative development and rapid feedback cycles. By creating tangible artifacts, designers can quickly gather feedback, test assumptions, and make necessary improvements. This iterative approach enables faster learning and refinement of ideas, leading to better-designed solutions.

v. Engages Stakeholders: Visual representations have a compelling impact on stakeholders and clients. Seeing tangible prototypes or visual designs helps them visualize the final product, elicits emotional responses, and promotes a sense of ownership and engagement. It enables stakeholders to provide more meaningful input and make informed decisions based on their direct interaction with the design.

Overall, adopting a show-don't-tell mindset in the design thinking process helps individuals effectively communicate ideas, collaborate, iterate, and engage stakeholders. It fosters a more inclusive and participatory approach to design, leading to better solutions and outcomes.

Q12. List any 3 business-oriented designs which are supported by IDEO. Also, write your observation on the design of GE Adventure series MRI machines as compared to Old MRI machines.

IDEO, a renowned design and innovation consultancy, has been involved in numerous business-oriented design projects.

Airbnb: IDEO worked closely with Airbnb to redesign and improve the user experience of their online platform. They focused on enhancing trust and making the booking process more intuitive and user-friendly. This collaboration helped Airbnb become a leading platform for peer-to-peer home-sharing.

Procter & Gamble: IDEO collaborated with Procter & Gamble on various projects, including the redesign of packaging for household products. They focused on creating more sustainable and environmentally friendly packaging designs while maintaining functionality and brand appeal.

Coca-Cola's Freestyle Fountain Dispenser: IDEO collaborated with Coca-Cola to design the Freestyle fountain dispenser, a self-serve machine that offers a wide variety of beverage choices. IDEO's design focused on enhancing the user experience, customization options, and operational efficiency.

Walmart's Sustainability Index: IDEO worked with Walmart to design the Sustainability Index, a tool that measures and evaluates the environmental impact of products. This design initiative aimed to promote sustainability and enable informed consumer choices.

The Design of Apple's Macintosh: IDEO played a significant role in the design of the original Macintosh computer in the 1980s. They collaborated with Apple to create a user-friendly interface, innovative hardware design, and a visually appealing product that revolutionized the personal computer industry.

GE Adventure series MRI machines vs Old MRI machines.

Diagnostic imaging procedures are cutting-edge technology, but at the same time they are an unpleasant experience for patients – and even more for pediatric patients. During the usage of old MRI machines, parents and kids were upset and filled with anxiety due to the upcoming procedure. The dark MRI room consisted of a warning sticker with an exclamation point, yellow and black caution tape on the floor, and the MRI machine that made a terrible noise and “looked like a brick with a hole in it.” Kids’ terrified reaction to these scanners has realized that there is a need to focus on making the stressful experience more pleasant and fun for MRI users. Designers at IDEO gained empathy with children by visiting a local daycare and talking

to pediatric healthcare professionals to find out what children went through as patients. They collaborated with a team of educators from a local children's museum and a group of medical professionals to produce ideas that could help make the scanning experience enjoyable.

Using his new understandings, Doug Dietz has designed the GE Adventure Series, themed scanning experiences that made medical procedures more like exciting adventures. For example, the Pirate Adventure transformed the medical setting into the dock of a pirate ship, where the patients are scanned on its plank. In the Coral City Adventure, patients in the emergency room go into a yellow submarine for their procedures, with the sound of harps in the background. In the Cozy Camp, patients are placed in a camping scene where they are scanned in a sleeping bag under the starry night. With these new magical settings and experiences, satisfaction scores from children rose up to 90 per cent, with some children asking parents if they could do the procedure again, demonstrating the power of empathy in insightful design. These companies are now being recognized by their customers because they are designing based on customer needs instead of gaining attention through advertisements.

Q13. Differentiate between “Growth Mindset” and “Beginner’s Mindset”.

In the context of design thinking, both the "Growth Mindset" and the "Beginner's Mindset" represent valuable perspectives that can foster creativity, innovation, and problem-solving. While there may be some overlap between the two, they differ in certain key aspects:

i. Growth Mindset:

Focus: The growth mindset emphasizes the belief that abilities and intelligence can be developed through dedication, effort, and learning. It emphasizes the idea that skills can be improved over time.

Adaptability: Individuals with a growth mindset are open to challenges and see failures or setbacks as opportunities for learning and growth. They embrace obstacles as chances to refine their skills and develop new strategies.

Resilience: Those with a growth mindset are persistent and motivated to overcome obstacles. They see setbacks as temporary and believe in their ability to improve and achieve success through hard work.

Continuous Improvement: A growth mindset encourages continuous learning and improvement. It promotes the idea of seeking feedback, learning from others, and constantly refining skills to reach higher levels of competence.

ii. Beginner's Mindset:

Openness: The beginner's mindset refers to approaching a situation or problem with a fresh perspective as if encountering it for the first time. It involves setting aside preconceived notions, biases, and assumptions.

Curiosity: Individuals with a beginner's mindset display a sense of curiosity, eagerness to explore, and a willingness to ask questions. They are open to new ideas and possibilities, without being constrained by past experiences or knowledge.

Unbiased Observation: The beginner's mindset encourages observing and experiencing things without judgment or prior expectations. It involves being fully present and attentive, enabling fresh insights and novel approaches to emerge.

Humility: Adopting a beginner's mindset involves recognizing that there is always more to learn and embrace the idea that expertise is not a barrier to innovation. It entails embracing the role of a learner, even for individuals with considerable experience.

Q14. Frame explorative, affective, reflective, probing and analytical questions to interview the target group of children aged between 12 and 15 towards the use of gaming toys.

1. What types of gaming toys do you currently own or use?
 2. How often do you engage with gaming toys in your free time?
 3. What are some of the reasons you enjoy playing with gaming toys?
 4. Are there any specific features or aspects of gaming toys that attract you the most?
 5. Do you prefer playing with gaming toys alone or with friends? Why?
 6. How do you feel when you play with gaming toys?
 7. Can you describe a memorable experience or feeling you've had while playing with a gaming toy?
 8. Do gaming toys help you relax, have fun, or feel a sense of achievement? Why or why not?
 9. How important are gaming toys in your overall entertainment or recreational activities?
 10. Do you have any emotional attachment to specific gaming toys? If so, why?
 11. Have gaming toys influenced your creativity or imagination? If yes, can you provide an example?
 12. Have you ever learned something new or developed a new skill while playing with gaming toys? Please elaborate.
 13. Can you think of any ways gaming toys have impacted your problem-solving abilities?
 14. Do gaming toys provide you with a sense of accomplishment or satisfaction? Why or why not?
 15. How do you perceive the role of gaming toys in your personal development?
 16. Can you explain how gaming toys differ from other types of toys you play with?
 17. What aspects of gaming toys could be improved to make them even more enjoyable or engaging for you?
 18. Are there any challenges or frustrations you face when using gaming toys? Can you provide specific examples?
 19. How do you decide which gaming toys to purchase or ask for as gifts?
 20. Are there any concerns or limitations you have regarding the use of gaming toys? Please explain.
 21. In your opinion, what are the benefits of playing with gaming toys?
 22. How do you think gaming toys influence your cognitive skills, such as problem-solving or strategic thinking?
 23. Can you identify any potential drawbacks or negative effects of the excessive use of gaming toys?
 24. Do you believe gaming toys have any educational value? Why or why not?
 25. How do you think gaming toys might evolve or change in the future?
-

Q15. “Having a human-centred mindset and being mindful of process help designers to use design thinking to its maximum potential.” Substantiate this statement with relevant explanations and examples.

The statement "Having a human-centred mindset and being mindful of process help designers to use design thinking to its maximum potential" is substantiated by the following explanations and examples:

1. Human-Centered Mindset:

Design thinking emphasizes the importance of understanding and empathizing with the end-users. By adopting a human-centred mindset, designers focus on the needs, desires, and experiences of the users, which leads to more effective and meaningful solutions.

a. Empathy: Designers with a human-centred mindset actively seek to understand the perspectives, emotions, and challenges of the users. This empathy allows them to gain insights into user behaviours, motivations, and preferences. For example, when designing a new healthcare app, a designer with a human-centred mindset might conduct user research, interviews, or surveys to understand the pain points and requirements of patients and healthcare professionals. This empathetic understanding helps the designer create a solution that truly addresses the user's needs.

b. User Feedback and Iteration: A human-centred mindset also involves actively involving users throughout the design process and incorporating their feedback. By soliciting user feedback at various stages, such as concept testing or usability testing, designers can refine their solutions iteratively. This feedback loop helps designers validate their assumptions, uncover usability issues, and make necessary improvements. For instance, a designer developing a new e-commerce website might create prototypes and gather user feedback to refine the user interface, navigation, and overall user experience based on the user's preferences and expectations.

2. Mindfulness of Process:

Being mindful of the design thinking process ensures that designers follow a structured approach, leverage each stage effectively, and maximize the potential for innovative solutions.

a. Problem Framing: The initial stage of design thinking involves clearly defining and understanding the problem. Designers who are mindful of this stage carefully analyze the context, user needs, and business goals. By thoroughly framing the problem, designers can identify the root causes, define the scope, and set specific objectives for their design process. For example, if a designer is tasked with improving the checkout process of an e-commerce platform, they need to understand the pain points of users, business requirements, and technological constraints to effectively frame the problem.

b. Ideation and Divergent Thinking: In the ideation stage, designers generate a wide range of ideas without judgment, encouraging creativity and innovation. Being mindful of this stage allows designers to explore various possibilities and alternatives. They can use brainstorming techniques, sketching, or collaborative workshops to foster divergent thinking. For instance, when designing a new educational game, designers may generate multiple ideas for game mechanics, educational content, and user interactions. By embracing divergent thinking, designers can discover novel and unexpected solutions.

c. Prototyping and Testing: Designers who are mindful of the prototyping and testing stages create prototypes or minimum viable products (MVPs) to gather feedback and validate their design assumptions. By building tangible representations of their ideas, designers can quickly test and refine their solutions based on user interactions and feedback. For example, a designer creating a mobile app for event planning might create a paper prototype or a clickable wireframe to simulate the app's interface and functionalities. User testing with the prototype can reveal usability issues, navigation challenges, or feature preferences that guide further iterations.

By combining a human-centred mindset with mindfulness of the design thinking process, designers can ensure that their solutions are deeply rooted in user needs, validated through feedback, and iteratively improved. This approach enables them to leverage design thinking to its maximum potential, resulting in innovative, user-centric, and impactful designs.

Q16. Why is it called 'phases' of design thinking instead of 'stages'?

The use of the term "phases" instead of "stages" in design thinking is primarily due to the iterative and non-linear nature of the process. Here are a few reasons why "phases" is commonly used:

- i. ***Iterative Process:*** Design thinking is an iterative process that involves cycling back and forth between different phases. Designers often revisit and iterate on earlier phases based on new insights and feedback. The term "phases" reflects the idea that design thinking is not a linear progression but a continuous exploration and refinement of ideas.
- ii. ***Flexibility and Adaptability:*** Design thinking encourages flexibility and adaptability, allowing designers to adjust their approach based on evolving needs and challenges. The term "phases" conveys a sense of fluidity and openness to change, as designers may need to move back and forth between different phases depending on the context and feedback received.
- iii. ***Non-Linear Problem-Solving:*** Design thinking acknowledges that problem-solving is rarely a linear process. It recognizes that different activities and perspectives may be needed simultaneously or at different points in the process. Using the term "phases" captures the non-linear nature of problem-solving in design thinking and emphasizes the need for exploration and iteration.
- iv. ***Creative Exploration:*** Design thinking emphasizes the importance of divergent thinking and generating a wide range of ideas. The term "phases" aligns with the idea that each phase represents a different aspect of the design process, fostering creative exploration and allowing designers to approach problem-solving from multiple angles.

Overall, using the term "phases" in design thinking emphasizes the iterative, flexible, and non-linear nature of the process. It acknowledges that design thinking is not a rigid step-by-step approach but a dynamic and adaptable framework for problem-solving and innovation.

Q17. Compare the design thinking approach for education and for business. Give suitable examples for both.

While the fundamental principles of design thinking apply to both education and business, the specific applications and objectives differ based on the context. In education, the focus is on enhancing teaching and learning, while in business, it is on driving innovation and improving customer experiences.

i. Design Thinking in Education:

Focus: In education, the design thinking approach is primarily focused on improving teaching and learning experiences, curriculum development, and student engagement.

User: The primary users in education are students and teachers. Design thinking aims to understand their needs, challenges, and aspirations to create effective and learner-centered solutions.

Example: Redesigning a Learning Experience

In education, design thinking can be applied to redesign a learning experience. For instance, a design thinking approach can be used to create a hands-on science project that engages students in active experimentation, critical thinking, and problem-solving. The process involves empathizing with students, ideating different project ideas, prototyping and testing the projects, and refining them based on student feedback. The goal is to create a meaningful and impactful learning experience that addresses the specific needs and interests of the students.

ii. Design Thinking in Business:

Focus: In business, the design thinking approach is often used to drive innovation, solve complex problems, and improve products, services, or processes. The focus is on understanding customer needs and creating user-centred solutions.

User: The users in business are customers or end-users of products or services. Design thinking aims to deeply understand their needs, pain points, and desires to develop innovative solutions that meet their expectations.

Example: Improving Customer Experience

In a business context, design thinking can be applied to improve the customer experience. For example, a hotel chain may use design thinking to enhance the check-in process. This involves empathizing with guests, identifying pain points in the existing process, brainstorming and prototyping solutions, and testing them with customers. The final solution may involve a mobile check-in app, streamlined procedures, and personalized service, all aimed at providing a seamless and delightful experience for the guests.

UNIT-II: The Design Thinking Process

Q1. Differentiate between finding and insight with examples.

In the context of design thinking and research, "finding" and "insight" are related but distinct concepts. Here's a differentiation between the two with examples:

i. Finding:

A finding refers to a specific piece of information or observation that is discovered through research or data analysis. Findings are factual and descriptive in nature, providing evidence or data points that contribute to the understanding of a particular problem or situation. Findings are often objective and can be quantifiable or qualitative in nature.

Example: In a user research study for a mobile banking app, a finding could be "75% of the participants reported difficulty navigating through the app's menu structure."

ii. Insight:

An insight, on the other hand, goes beyond the mere observation of findings. It involves a deeper understanding and interpretation of the findings, uncovering underlying patterns, motivations, or hidden needs. Insights provide a meaningful and actionable understanding that can inspire design solutions and drive innovation.

Example: Building on the finding mentioned earlier, insight could be "Users find it frustrating to locate specific features in the app's menu due to inconsistent labelling and lack of clear visual hierarchy. They desire a streamlined and intuitive navigation system that aligns with their mental model of a banking app."

In summary, findings are the raw data or observations collected during research, while insights are the meaningful interpretations or understandings derived from those findings. Insights provide designers with a deeper understanding of users' needs, desires, and pain points, leading to more effective and human-centred design solutions.

OR

i. Findings:

Findings are the raw data or information that is discovered through research or analysis. They are factual, and descriptive, and provide specific observations or measurements. Findings are typically the initial discoveries made during the research process.

Example:

In a survey conducted to understand consumer preferences for a new beverage product, the findings could include:

- 60% of respondents prefer a sweeter taste.
- 70% of respondents prioritize convenience in packaging.
- 45% of respondents consume beverages primarily for hydration.

ii. Insights:

Insights go beyond the surface-level observations of findings. They involve a deeper understanding, interpretation, and synthesis of the data. Insights provide meaningful and actionable understanding that helps answer the "why" behind the findings. They reveal underlying motivations, patterns, or unmet needs.

Example:

Building upon the findings mentioned earlier, insights could be:

- Consumers' preference for sweeter taste indicates a demand for indulgent and flavorful beverage options.
- The emphasis on convenience in packaging suggests a need for on-the-go consumption and ease of use.
- Consumers' prioritization of hydration highlights an opportunity to promote the product's health and wellness benefits.

In this example, the findings provide specific data points, while the insights offer an interpretation of the findings and uncover the underlying motivations and opportunities for product development and marketing strategies.

To summarize, findings are the raw data or observations, while insights are the meaningful interpretations and understanding derived from those findings. Insights provide deeper understanding, guide decision-making, and inspire actions or design solutions.

Q2. Identify the error in the following actionable problem statement, and comment on the same. “Working Professionals need a way to rest and relax”

The error in the given actionable problem statement is that it lacks specificity and clarity. The problem statement does not provide enough details about the specific needs and context of the working professionals. It is too broad and general, making it difficult to develop focused solutions.

To improve the problem statement, it should be more specific and address the specific pain points or challenges faced by working professionals when it comes to resting and relaxing. This could involve understanding their constraints, preferences, and any specific factors that hinder their ability to rest and relax effectively. By identifying the specific needs and challenges, it becomes easier to develop targeted solutions that truly address the problem at hand.

A revised and more actionable problem statement could be: "Working professionals in high-stress industries need a convenient and time-efficient way to unwind and reduce stress during their limited downtime." This revised statement focuses on the target audience (working professionals in high-stress industries), highlights their specific challenges (limited downtime), and emphasizes the need for a convenient and time-efficient solution. This provides a clearer direction for designing solutions that effectively meet the needs of the target group.

Q3. Illustrate the Empathy map with a diagram.

Name: _____ Date: _____

EMPATHY MAP FOR STUDENTS	
1. Who Who are we empathizing with? What is the situation they are in?	2. Need to Do What do they need to do? What do they need to do differently? What job(s) do they want or need to get done? What decision(s) do they need to make? How will we know they were successful?
3. See What do they see? What do they see in their environment? What are they watching and reading?	4. Say What do they say? What can we imagine them saying?
5. Do What do they do? What actions and behaviors did you notice? What did they do today? What can you imagine them doing?	6. Hear What do they hear? What do they hear others say? What did they hear secondhand?
7. Think and Feel What are the thoughts and feelings that motivate their behavior? What are their fears, frustrations, and anxieties? What are their wants, needs, hopes, and dreams?	
Finally, find <u>Needs</u> and <u>Insights</u> of your user.	

Q4. “To empathize with others, you need to empathize with yourself” Justify the statement.

The statement "To empathize with others, you need to empathize with yourself" highlights the importance of self-awareness and understanding in developing empathy towards others. Here are a few justifications for this statement:

1. **Understanding Emotions:** By empathizing with ourselves, we gain a deeper understanding of our own emotions, needs, and experiences. This self-awareness allows us to recognize and connect with similar emotions and experiences in others, enabling us to empathize more effectively.
2. **Recognizing Bias:** Self-reflection helps us become aware of our own biases, assumptions, and preconceived notions. When we understand our own biases, we can work towards overcoming them and approaching others with an open mind, facilitating better empathy.
3. **Developing Compassion:** When we practice self-compassion and self-empathy, we cultivate a compassionate mindset towards ourselves. This compassion can then extend to others, allowing us to connect with their struggles, challenges, and joys in a more authentic and empathetic way.
4. **Building Resilience:** Understanding our own emotions and experiences helps us develop resilience and emotional intelligence. This resilience enables us to empathize with others during difficult situations, as we can draw from our own experiences and emotions to provide support and understanding.

5. Enhancing Communication: Empathizing with ourselves improves our communication skills, as it enables us to express our thoughts and emotions more effectively. This enhanced communication ability allows us to connect with others on a deeper level, fostering stronger empathy and understanding.

Ultimately, the process of empathizing with oneself is a fundamental aspect of developing empathy towards others. It helps us develop self-awareness, recognize biases, cultivate compassion, build resilience, and enhance communication skills. By understanding our own emotions and experiences, we can better relate to and connect with others, leading to more meaningful and genuine empathy.

Q5. What are the strategies listed to help one to 'listen and not judge' during empathize phase? Why is this mindset important?

To help one "listen and not judge" during the empathize phase of design thinking, several strategies can be employed:

- 1. Practice Active Listening:** Focus on fully understanding the speaker by giving them your undivided attention. Avoid interrupting or formulating responses in your mind while the speaker is talking. Instead, actively listen to their words, tone, and non-verbal cues.
- 2. Suspend Assumptions and Biases:** Set aside preconceived notions and biases about the user or the problem at hand. Approach the conversation with an open mind and be willing to challenge your assumptions.
- 3. Cultivate Empathy:** Put yourself in the shoes of the user and strive to understand their experiences, emotions, and perspectives. Try to see the world from their point of view and genuinely empathize with their situation.
- 4. Refrain from Judgments:** Avoid making quick judgments or forming opinions about the user or their experiences. Recognize that everyone has unique perspectives and that judgments can hinder your ability to understand their needs and motivations.
- 5. Ask Open-Ended Questions:** Encourage the user to share their thoughts, feelings, and experiences by asking open-ended questions. These questions invite deeper reflection and provide an opportunity for the user to express themselves freely.

This mindset of "listening and not judging" is crucial during the empathize phase for several reasons:

- 1. Encourages Open Communication:** When users feel that they are being heard without judgment, they are more likely to share their honest thoughts, experiences, and needs. This open communication allows designers to gather more accurate and meaningful insights.
- 2. Enables Deeper Understanding:** By suspending judgment, designers can focus on truly understanding the user's perspective. This deeper understanding helps uncover hidden insights and unmet needs that may have been overlooked if quick judgments were made.
- 3. Builds Empathy and Trust:** When users feel listened to and understood, it builds a sense of empathy and trust between the designer and the user. This trust fosters a more collaborative and open relationship, which can lead to better design solutions.

4. Facilitates User-Centered Design: Design thinking aims to create solutions that meet the needs of the users. By listening without judgment, designers can gather insights that form the basis for user-centered design, resulting in solutions that truly address user needs and desires.

In summary, adopting a "listen and not judge" mindset during the empathize phase allows designers to create a safe and open space for users to share their experiences. It helps foster empathy, build trust, and gather deep insights that inform the subsequent stages of the design process, leading to more meaningful and impactful design solutions.

Q6. Identify the target user market for the following brands, and explain briefly why you chose that group as the target user market. (i) Whiteboards, (ii) Wireless computer keyboard, (iii) Water spray can, (iv) nylon rope.

(i) Whiteboards:

Target User Market: Educational institutions, offices, and collaborative workspaces.

Explanation: Whiteboards are commonly used in educational institutions such as schools and universities, as well as in offices and collaborative workspaces. These spaces require a tool for visualizing ideas, brainstorming, and facilitating group discussions. Whiteboards offer a convenient and reusable surface for writing and erasing content, making them ideal for collaborative learning and sharing information in a group setting.

(ii) Wireless Computer Keyboard:

Target User Market: Professionals, remote workers, and tech-savvy individuals.

Explanation: The target user market for wireless computer keyboards consists of professionals, remote workers, and tech-savvy individuals. These individuals often work on computers for extended periods and require a comfortable and efficient typing experience. Wireless keyboards provide flexibility and convenience by eliminating the need for tangled cords and allowing users to work from a distance. Professionals and remote workers appreciate the portability and ease of use offered by wireless keyboards, while tech-savvy individuals value the latest technology and enjoy the convenience of a cable-free setup.

(iii) Water Spray Can:

Target User Market: Gardeners, plant enthusiasts, and household cleaning purposes.

Explanation: The target user market for water spray cans includes gardeners, plant enthusiasts, and individuals who use them for household cleaning purposes. Gardeners and plant enthusiasts utilize water spray cans to provide a controlled and precise watering method for plants. The spray nozzle allows for gentle watering, avoiding damage to delicate plants. Additionally, water spray cans are useful for household cleaning tasks such as misting indoor plants, dampening fabrics, or refreshing surfaces.

(iv) Nylon Rope:

Target User Market: Outdoor enthusiasts, campers, hikers, and DIY enthusiasts.

Explanation: The target user market for nylon rope consists of outdoor enthusiasts, campers, hikers, and DIY enthusiasts. Nylon rope is highly durable and versatile, making it suitable for a wide range of applications in outdoor activities such as camping, hiking, and climbing.

Outdoor enthusiasts rely on nylon rope for tasks like securing tents, creating clotheslines, setting up hammocks, or making DIY repairs. Its strength, flexibility, and resistance to abrasion make it a preferred choice for various outdoor and DIY applications.

Q7. Name and explain any five titles under which the findings are classified in the empathy map.

The Empathy Map is a graphic organizer that visually displays and classifies the information captured during empathy work. During interviews, observations, and immersion, students record their findings in their map and classify them into seven empathic areas of focus, all in the perspective of the user. Created by Dave Gray, the founder of XPLANE, a consulting company, the Empathy Map was originally a business toolkit to help organizations gain a better understanding of their customers and their experiences. However, this tool is a great way for students to organize their findings, consider the motivations behind observed actions, and easily identify needs and insights.

1. *WHO are we empathizing with?* In this section, students identify the person or group of people they are designing for, the context of the design challenge, and the role of the user(s) in the context.

2. *What do they NEED to do?* Students list what their users need and want in regard to the design challenge. This includes tasks that their users want to accomplish and decisions they want to make. Additionally, students create criteria of what their users should be able to do to be successful in the design challenge. This section brings clarity and purpose on what they are trying to do to help their user(s).

3. *What do they SEE?* These findings are based on what the user sees in the context of the design challenge, in their physical environment, as well as outside of these areas that might be influential to them.

4. *What do they SAY?* Students record what they hear their user say that might be influential to the design process, as well as what they imagine their user saying.

5. *What do they DO?* Students identify all the things that their user does in the context of the design challenge and in their daily life. They can also list some behaviors that they imagine their users to do.

6. *What do they HEAR?* These findings are based on what the user(s) hear from others, the people around them, or what they hear secondhand.

7. *What do they THINK and FEEL?* Students list what their user(s) fear, are frustrated with and anxious about, as well as what they need, want, and hope for.

According to Dave Gray, the first two questions identify the goal of the activity and clarify the context of the design challenge. The following four questions focus on the observable experiences of the user that allow students to walk “a mile in their shoes.” The final question is considered one of the most important parts of the map because it requires students to imagine and find out what their user is thinking or feeling—the most significant action one can take to empathize with someone.

Q8. What are the three elements of the problem statement?

The three elements of a problem statement in design thinking are:

1. User: This element focuses on identifying and understanding the target user or customer for whom the problem needs to be solved. It involves gaining empathy and deep insights into the user's needs, desires, behaviours, and pain points. By clearly defining the user, design thinking ensures a user-centred approach to problem-solving.

2. Need or Challenge: This element defines the specific need, challenge, or opportunity that requires attention and resolution. It is crucial to clearly articulate the problem or opportunity statement to ensure a focused and actionable problem-solving process. This element helps frame the problem in a way that guides the design thinking process towards generating innovative and effective solutions.

3. Insight: This element refers to the deep understanding gained from observing and empathizing with the user. Insights are the discoveries or observations about the user's needs, motivations, behaviours, and pain points. They go beyond surface-level observations and help uncover underlying problems or opportunities that can guide the design process. Insights play a critical role in generating creative ideas and designing solutions that address the core needs and desires of the user.

By considering these three elements in the problem statement, design thinking ensures a human-centred and solution-focused approach, driving innovation and effective problem-solving.

Q9. Identify the target user for waterproof coating, and explain briefly why you choose that group.

The target user for waterproof coating can vary depending on the specific context and application. However, one potential target user group for waterproof coatings is outdoor enthusiasts and adventurers, such as hikers, campers, and backpackers.

Here's a brief explanation for choosing this group:

1. Relevance: Outdoor enthusiasts engage in activities that often expose them to various weather conditions, including rain, snow, and humidity. They rely on gear and equipment that need to withstand these elements to ensure their safety, comfort, and overall experience. Waterproof coatings can be beneficial for protecting their gear, such as tents, backpacks, jackets, and footwear, from moisture and extending their lifespan.

2. Need for Performance: Outdoor enthusiasts depend on their gear to perform at its best in challenging environments. Waterproof coatings can enhance the performance of their equipment by providing a protective barrier against water, preventing it from seeping through and compromising functionality. This group requires reliable and durable waterproofing solutions to maintain their gear's performance in adverse conditions.

3. Market Demand: The outdoor industry is thriving, and there is a growing market for high-quality outdoor gear and equipment. Waterproof coatings cater to the needs and demands of outdoor enthusiasts who prioritize durability, weather resistance, and longevity in their gear. Targeting this group allows manufacturers and suppliers of waterproof coatings to tap into a niche market with specific requirements and preferences.

It's important to note that the target user group for waterproof coatings can extend beyond outdoor enthusiasts to include professionals working in industries where water resistance is crucial, such as construction workers, maintenance personnel, and people involved in water-related activities. The choice of the target user group ultimately depends on the specific product, its intended use, and the market segment that offers the most potential for success.

Q10. What is the importance of the ‘how might we’ question?

The "How Might We" (HMW) question is a critical component of the design thinking process. It plays a crucial role in framing the problem and inspiring creative solutions. Here are the key reasons why the HMW question is important:

- 1. Encourages Ideation:** The HMW question prompts brainstorming and ideation. It invites designers and problem solvers to think broadly and generate a wide range of ideas and possibilities. By framing the question in an open-ended and optimistic manner, it sparks creative thinking and opens up new avenues for exploration.
- 2. Shifts the Mindset:** The HMW question shifts the focus from the problem itself to potential solutions. It helps to reframe challenges as opportunities for innovation. By asking "How might we...", it assumes that solutions exist and invites individuals to approach the problem with a positive and solution-oriented mindset.
- 3. Promotes Collaboration:** The HMW question fosters collaboration and teamwork. When presented with the question, individuals from diverse backgrounds and perspectives can come together to generate ideas and build upon each other's contributions. It creates a shared understanding of the problem and encourages collective problem-solving.
- 4. Sparks Creativity:** The HMW question stimulates creative thinking by encouraging individuals to think outside the box. It gives permission to explore unconventional and innovative solutions. By removing constraints and limitations, it opens up possibilities for unique and breakthrough ideas.
- 5. Guides the Design Process:** The HMW question serves as a guiding principle throughout the design process. It helps maintain a clear focus on the problem and ensures that ideation and prototyping efforts are aligned with addressing the identified challenge. It provides a sense of direction and purpose as designers navigate through the various stages of the design thinking process.

Overall, the HMW question is a powerful tool for framing the problem and generating innovative solutions. It sets the stage for creative ideation and guides the design process towards meaningful and impactful outcomes.

Q11. Write ten interview questions to ask students of rural areas who have problems pertaining to education through online classes.

Here are ten interview questions you can ask students from rural areas who are facing challenges with online education:

1. How has the transition to online classes affected your learning experience?
2. What specific challenges do you face in accessing and using technology for online learning?
3. Can you describe the quality of your internet connectivity in your area? Are there any issues or limitations you encounter?

4. What difficulties do you encounter in understanding and following online instructions or course materials?
5. Are there any financial constraints or lack of resources that hinder your ability to participate in online classes effectively?
6. How do you manage distractions and maintain focus during online classes, considering the environment you study in?
7. Do you feel comfortable asking questions or seeking clarification from teachers during online classes? If not, what prevents you from doing so?
8. Are there any language or cultural barriers that make it challenging for you to fully engage in online learning?
9. How do you stay motivated and engaged in online classes without the physical presence of teachers and classmates?
10. What additional support or resources do you think would be helpful for you to overcome the challenges you face with online education?

Q12. What is the synthesis phase in design thinking? What does one do during this phase?

The synthesis phase in design thinking is a critical step in the design process that involves analyzing and making sense of the data, insights, and observations gathered during the research and empathy phase. It is the stage where the design team collaboratively interprets and synthesizes the information to identify patterns, themes, and opportunities for design.

During the synthesis phase, the design team typically engages in activities such as:

- 1. Data Analysis:** The collected data, including interview transcripts, observation notes, and survey responses, is reviewed and analyzed to identify key findings, trends, and commonalities. The team looks for patterns, outliers, and connections between different data points.
- 2. Affinity Mapping:** The team uses techniques like affinity mapping to group related data and insights together. Post-it notes or digital tools are often used to capture individual observations and then rearranged into clusters based on common themes or ideas. This process helps organize and visualize the data in a meaningful way.
- 3. Insights Generation:** The team identifies insights by extracting meaning from the synthesized data. Insights are often actionable statements that capture the underlying needs, desires, and pain points of the users. These insights provide a deeper understanding of the problem space and serve as a foundation for ideation and solution generation.
- 4. Persona Creation:** Based on the synthesized data, the team may develop user personas. Personas are fictional representations of the target users, created to embody the characteristics, motivations, and behaviors of specific user segments. Personas help the team gain a human-centered perspective and guide the design process towards meeting user needs.
- 5. Problem Framing:** The team refines and frames the problem statement based on the insights and understanding gained during the synthesis phase. A well-defined problem statement sets the direction for ideation and solution development.
- 6. Storytelling:** The team may use storytelling techniques to communicate the synthesized findings and insights effectively. This could involve creating narratives, user journey maps, or experience maps that illustrate the user's current situation, pain points, and opportunities for improvement.

Overall, the synthesis phase is about distilling large amounts of data and observations into meaningful insights that inform the subsequent design steps. It is a collaborative and iterative process that requires the design team to make sense of the research findings, identify patterns, and reframe the problem statement to guide the ideation and prototyping phases.

Q13. What is an empathy map? What are its uses?

An empathy map is a visual tool used in design thinking to gain a deeper understanding of users or stakeholders. It helps designers develop empathy by capturing their thoughts, feelings, behaviours, and needs.

The key components of an empathy map typically include the following:

1. User: The person or group being studied is at the center of the empathy map. This could be a specific user persona or a broader target audience.
2. Says: This section captures the explicit statements or quotes from the user, providing insights into what they say and express verbally.
3. Thinks: This section delves into the user's thoughts, beliefs, and assumptions. It helps uncover their underlying motivations, values, and priorities.
4. Feels: This section explores the user's emotions, both positive and negative. It includes their fears, desires, frustrations, and aspirations, providing a deeper understanding of their emotional state.
5. Does: This section focuses on the user's observable behaviors and actions. It helps identify their habits, routines, and interactions related to the topic of study.

The uses of an empathy map are as follows:

1. User Understanding: An empathy map enables designers to gain a comprehensive understanding of the user's needs, motivations, and pain points. It helps develop a user-centred perspective.
2. Empathy Development: By considering the user's thoughts, feelings, and behaviours, an empathy map promotes empathy among designers. It encourages a deeper connection with the user's experiences and challenges.
3. Ideation and Problem-Solving: The insights gathered from an empathy map serve as a foundation for generating innovative ideas and solutions. It sparks creativity and guides the design process towards addressing user needs.
4. Stakeholder Alignment: An empathy map is a valuable communication tool to share user insights with stakeholders, team members, and clients. It fosters shared understanding and collaboration in designing effective solutions.
5. Decision-making: An empathy map provides designers with a framework for making informed design decisions. It helps prioritize features, functionalities, and design choices based on user needs and motivations.

Q14. Write ten interview questions to ask hospital staff pertaining to the issues regarding the disposal of gloves and needles.

Here are ten interview questions to ask hospital staff regarding the disposal of gloves and needles:

1. How are gloves and needles currently disposed of in our hospital?
2. What are the main challenges or difficulties you face when it comes to the proper disposal of gloves and needles?
3. Are there any specific guidelines or protocols in place for the disposal of gloves and needles? If so, can you explain them?
4. What safety measures are taken to ensure the proper disposal of gloves and needles to protect both staff and patients?
5. Have there been any incidents or concerns related to the improper disposal of gloves and needles that you are aware of?
6. Are there any environmental considerations or regulations that impact the disposal of gloves and needles?
7. How are hospital staff educated and trained on the proper procedures for disposing of gloves and needles?
8. Are there any specific challenges or limitations related to the disposal process that you would like to address or improve?
9. Are there any cost implications or resource constraints that affect the disposal practices for gloves and needles?
10. Can you suggest any ideas or improvements that could enhance the disposal process for gloves and needles in our hospital?

These questions can help gather valuable insights from hospital staff regarding the current practices, challenges, and potential areas for improvement in the disposal of gloves and needles.

Q15. The pottery industry of Majuli is characterized by the small size of units, family-based operation, use of traditional tools and equipment, landless people and illiteracy among the artist. The survey reveals that the pottery industry in Majuli suffered lots of problems due to irregular supply and absence of raw materials, financial support, outdated tools and equipment, competition from the large-scale sector, poor transport and lack of advertisement, absence of training workshops and lack of resource and development efforts. The promotion of the pottery industry is essential for the development of economic as well as cultural preservation. The intermediaries get good profits but they face the problem of getting the products damaged, demand dependent on the festive occasion, problems of transportation, etc. Study shows that there is a huge demand for Majuli pottery products in the international market. Buyers are willing to pay a premium for handmade goods, but the potters don't know how to initiate exports. Write one problem statement based on the above observation, and change it into a 'How Might We' question.

Problem Statement:

The pottery industry in Majuli faces multiple challenges, including irregular supply of raw materials, lack of financial support, outdated tools and equipment, competition from the large-scale sector, poor transport infrastructure, and inadequate promotion and development efforts. Despite a high demand for Majuli pottery products in the international market, the potters lack

the knowledge and resources to initiate exports, resulting in missed opportunities for economic growth and cultural preservation.

"How Might We" Question:

How might we empower and support the potters in the Majuli pottery industry to address the challenges of raw material availability, financial support, modernizing tools and equipment, overcoming competition, improving transportation infrastructure, promoting their products effectively, and providing training and resources to initiate successful exports, thereby enabling them to capitalize on the international demand for Majuli pottery and contribute to the economic and cultural development of the region?

Q16. What sort of research is necessary during the empathy phase, and why?

During the empathy phase in design thinking, qualitative research methods are typically used to gain a deep understanding of users, their needs, and their experiences. The research conducted during this phase aims to gather rich insights and generate empathy towards the users. Here are some key research methods used during the empathy phase:

1. User Interviews: Conducting interviews with users allows designers to gather in-depth information about their perspectives, behaviors, and challenges. Open-ended questions are used to explore their needs, motivations, and goals, providing valuable insights into their experiences.

2. Observation: Observing users in their natural environment helps designers gain a firsthand understanding of their behaviors, interactions, and pain points. By observing how users navigate their daily tasks and activities, designers can uncover valuable insights and identify opportunities for improvement.

3. Immersion and Contextual Inquiry: Immersing oneself in the users' environment and engaging in contextual inquiry involves experiencing and understanding their world firsthand. Designers participate in the users' activities, interact with their surroundings, and observe the challenges they face. This approach helps designers gain deeper insights into the context in which users operate and develop a more empathetic perspective.

4. Empathy Interviews: Empathy interviews go beyond collecting factual information and aim to uncover users' emotions, desires, and motivations. By creating a safe and open space, designers encourage users to share their feelings, experiences, and aspirations. This helps designers empathize with users and develop a deeper understanding of their needs.

5. Empathy Tools and Techniques: Designers may use various tools and techniques to support empathy building, such as empathy maps, personas, and journey maps. These tools help synthesize research findings and create visual representations that aid in understanding and communicating user experiences.

The research conducted during the empathy phase is crucial for several reasons:

1. Understand User Needs: It helps designers gain a deep understanding of users' needs, challenges, and aspirations. By uncovering these insights, designers can identify opportunities to create meaningful and impactful solutions.

2. Foster Empathy: Research during the empathy phase allows designers to develop empathy towards users by truly understanding their experiences, emotions, and perspectives. This empathy helps designers design more human-centred solutions.

3. Challenge Assumptions: It helps challenge preconceived notions and biases that designers may have about users. By conducting research, designers can validate or disprove assumptions, leading to more accurate and effective problem-solving.

4. Inform Design Decisions: The insights gained during the empathy phase inform subsequent phases of the design process, such as ideation and prototyping. They guide designers in generating ideas that directly address user needs and preferences.

Overall, research during the empathy phase is essential for design thinking as it forms the foundation for user-centred design. It ensures that the design solutions are based on a deep understanding of users and their needs, leading to more successful and impactful outcomes.

Q17. You are given the task of conducting an empathy interview for a group of new mothers regarding the design of a baby cradle. Form two empathy questions for each block of the empathy map, and create an empathy map.

User: New Mothers

1. Empathy Questions for "Says" (What the user says):

- What are the key features or characteristics you look for when choosing a baby cradle?
- Can you describe any specific challenges or concerns you have faced while using a baby cradle?

2. Empathy Questions for "Thinks" (What the user thinks):

- What are your thoughts about the safety features of baby cradles currently available in the market?
- Do you have any concerns about the comfort and support provided by existing baby cradles?

3. Empathy Questions for "Feels" (What the user feels):

- How do you feel when you place your baby in a cradle? Are there any emotions or worries that arise?
- Can you share your feelings regarding the aesthetic appeal and design of baby cradles?

4. Empathy Questions for "Does" (What the user does):

- Can you walk me through your daily routine involving the use of a baby cradle?
- How do you interact with your baby while they are in the cradle? What activities or tasks do you perform?

Empathy Map for New Mothers and the Design of a Baby Cradle:

"Says"	"Thinks"
"Feels"	"Does"

Q18. List the key rules and principles to be followed by an interviewer.

In the context of design thinking, here are key rules and principles to be followed by an interviewer:

- 1. Maintain a human-centred approach:** Design thinking emphasizes understanding the needs, desires, and experiences of users. As an interviewer, prioritize the user's perspective and aim to deeply understand their thoughts, feelings, and behaviours.
- 2. Cultivate empathy:** Design thinking relies on empathy to gain insights and develop meaningful solutions. Show empathy towards the interviewee by actively listening, demonstrating understanding, and validating their experiences.
- 3. Ask open-ended questions:** Frame questions that encourage the interviewee to share their stories, experiences, and perspectives in detail. Open-ended questions promote a deeper understanding and uncover insights that can drive the design process.
- 4. Foster a safe and inclusive environment:** Create a safe and inclusive space where the interviewee feels comfortable expressing their thoughts and opinions. Respect their diversity, backgrounds, and perspectives, and avoid judgment or bias.
- 5. Practice active listening:** Engage in active listening by giving the interviewee your full attention, maintaining eye contact, and showing genuine interest in their responses. Avoid interrupting and allow them to fully express themselves.
- 6. Avoid leading or biased questions:** Formulate questions that are neutral and unbiased, avoiding any influence towards a particular response. This helps to uncover genuine insights and prevents steering the interviewee's answers.
- 7. Be adaptable and flexible:** Design thinking requires flexibility and adaptability to respond to the unexpected. Be open to exploring new avenues and following tangents that may lead to valuable insights during the interview.
- 8. Stay focused on the user's experience:** Keep the interview centered around the user's experiences, needs, and challenges. Avoid making assumptions or bringing personal biases into the conversation.
- 9. Document and synthesize findings:** Take detailed notes or record the interview with the interviewee's permission. Capture key insights, observations, and quotes that can later be synthesized and analyzed to inform the design process.

10. Collaborate and iterate: Design thinking encourages collaboration and iteration. Involve other team members or stakeholders in the interview process to gather diverse perspectives. Iterate on the interview process itself, learning from each session to continually improve.

Q19. What are the three elements of a problem statement? Give an example.

The three elements of a problem statement in design thinking are:

1. User: This element focuses on identifying and understanding the target user or customer for whom the problem needs to be solved. It involves gaining empathy and deep insights into the user's needs, desires, behaviours, and pain points. By clearly defining the user, design thinking ensures a user-centred approach to problem-solving.

2. Need or Challenge: This element defines the specific need, challenge, or opportunity that requires attention and resolution. It is crucial to clearly articulate the problem or opportunity statement to ensure a focused and actionable problem-solving process. This element helps frame the problem in a way that guides the design thinking process towards generating innovative and effective solutions.

3. Insight: This element refers to the deep understanding gained from observing and empathizing with the user. Insights are the discoveries or observations about the user's needs, motivations, behaviors, and pain points. They go beyond surface-level observations and help uncover underlying problems or opportunities that can guide the design process. Insights play a critical role in generating creative ideas and designing solutions that address the core needs and desires of the user.

By considering these three elements in the problem statement, design thinking ensures a human-centred and solution-focused approach, driving innovation and effective problem-solving.

Example Problem Statement:

User: Small business owners in the food industry

Problem/Challenge: Limited visibility and reach of their products to potential customers

Context/Scope: Local community or neighbourhood

Revised Problem Statement:

"Small business owners in the food industry in a local community need a solution to increase the visibility and reach of their products to potential customers."

In this example, the problem statement includes all three elements. The user is small business owners in the food industry, the problem/challenge is the limited visibility and reach of their products, and the context/scope is the local community or neighbourhood. This problem statement provides a clear focus and sets the direction for finding solutions that address the specific needs of the target users in a defined context.